
Project: Loan Repayment

Name: Kolos Nagy

Date: 4/27/2026

Executive Summary (Maximum: One Page)

The purpose of the report is to analyze the key variables that affect loan repayment performance based on the data set provided. The report evaluates the relationship between the borrower's information to the attributes of the loans and their repayment status to identify the risk indicators. The goal of this analysis is to provide data driven insights for future reference and optimize lending strategies to ensure stability.

Key Findings: The variables for this analysis are interest rates, annual income, loan terms, loan amount, and employment length. The most significant factors are interest rate, annual income, and term to determine if the loan will become fully paid. The regression highlights that higher interest rates decrease the probability of loan repayment but on the other hand higher annual income and shorter terms will support the possibility of a fully paid loan status. The only variable that became insignificant was loan amount, meaning that the other variables are more important than the amount borrowed.

Recommended Course of Action: Based on these findings, it is advised to prioritize low risk borrowers with high annual income and short period of repayment. Encourage customers to take shorter term loans compared to long ones since they show better repayment ratios. Creating a program that could support lower income individuals with longer terms and lower interest rates, but they would need approval first.

Supporting Evidence: Loans that were Charged Off had a significantly higher average interest rate of 13.90% compared to 11.76% for those Fully Paid. Term Success Rates: 36-month loans achieved an 89.1% success rate, whereas 60-month loans showed a much lower repayment rate of 77.3%. Annual income and interest rates were confirmed as highly significant predictors, while loan amount failed to reach statistical significance, proving that the cost of debt is more critical than the amount of debt.

Section 1. Dependent and Independent Variables

Dependent Variable: Loan_status

Independent Variables:

- annual_income
- loan_amount
- interest_rate
- employment_length
- term

1. Annual Income: Positive relationship. It represents the borrower's ability to repay the loan. Borrowers with higher annual income generally are more financially stable and can make loan payments on time.

2. Loan Amount: Negative relationship. This is the total number that was borrowed. Larger loans usually require higher monthly payments, which can increase the pressure. Borrowers with higher loan amounts might struggle to repay them.

3. Interest Rate: Negative relationship. The interest rate determines how much additional money the borrower must pay on top of the original loan. Higher interest rates will increase monthly payments and the total. People with high rates might find it more difficult to repay loans.

4. Employment Length: Positive relationship. Employment length represents job stability and consistent income of the borrower. People employed for multiple years are usually more financially stable than the ones with less length. Longer employment could mean the loan will be paid with no problems.

5. Loan Term: Negative relationship. This refers to the length of time allowed to repay the loan. Longer loan terms reduce monthly payments but extend the duration which could lead to financial changes. Borrowers with longer terms might have more difficulty to repay than the ones with short terms.

Section 2. Data Transformation

Before starting the regression analysis, we had to ensure all the data is cleaned and good for the project. First, deleting all the values after the line that said: “Loans do not meet credit policy.”

Handling Missing Values and Errors

The process of cleaning the dataset started with dealing with the missing values issue. For this purpose, I used filters to detect any records that contained blanks, empty fields, or N/A's. This step is important because a regression cannot be calculated with incomplete observations, so I deleted all the missing records from the dataset to avoid any biases or calculation problems.

After I removed the blanks, I checked any records containing impossible values or data entry errors that couldn't have been removed with the missing values. Any error I found in the dataset was removed to maintain a clean sample.

Managing Outliers

Outliers can greatly affect regressions leading to misleading coefficients. To identify these, I used descriptive analysis and any observation that would fall outside the normal expectations were investigated. Finding huge numbers that created right-skewness I used logarithmic transformation to compress the scale of the data. This step helped normalize the data more than just dividing it and cleaning the data for the regression more.

Transformation of Non-Numeric Data

The regression analysis can only be done using numeric data. As such, all my categorical data was changed to numeric form. This was accomplished using dummy coding, in which categorical data were represented as zeros and ones. In addition, all data forms were checked for any numerical figures written in text.

After all the data cleaning was done, I did a final check of the descriptive statistics. Checking through the mean, median, standard deviation, skewness it was clear that there are no missing values, ranges are reasonable, and all the data is good for the analysis.

Section 3. Descriptives Statistics

Variables	Minimum	Maximum	Mean	Median
Loan_status	0	1	0.8594	1
Annual_Income	8.2940	15.6072	10.9854	11.0020
Loan_Amount	6.2146	10.4631	9.0966	9.2103
Interest_Rate	0.0542	0.2459	0.1205	0.1186
Emp_Lenght	0	10	4.9743	4
Term	0	1	0.7307	1

Loan_status (Dependent Variable): I transformed this categorical outcome into a binary dummy variable (0 and 1). With a mean of 0.859, the distribution indicates that approximately 86% of the observations are "1".

Annual_Income (Independent Variable): Because of "huge numbers" I applied a log transformation to annual income to make the minimum and maximum values closer together (Min = 8.29 Max = 15.61).

Loan_Amount (Independent Variable): To handle the large numerical scale of the data, I used a log transformation.

Interest_Rate (Independent Variable): I converted this percentage data into numeric. I verified that the data appears realistic without any impossible values like negative or extremely large values.

Emp_Lenght (Independent Variable): To convert the text data into numeric variables, I deleted all characters except for the numbers. In addition, I deleted all blank cells and kept the remaining data as years of employment (Range: 0-10).

Term (Independent Variable): I decided to change this variable into a dummy variable 0 and 1. By transforming the loan term (36 or 60 months) into binary form, the model will be able to recognize the effect of each loan period.

Section 4. Analysis

Pivot Tables:

The first pivot table I did was Term and Loan Status to see how term length affects the status.

Both variables used binary form:

- Term: 1 = 36 months 0 = 60 months
- Loan Status: 1 = Fully Paid 0 = Charged Off

After a successful pivot table analysis, it clearly shows us the results. 36-month loans make up the majority of the dataset with a high success rate of fully paid loans. Out of 28,287 observations 25,214 (89.1%) were fully paid and 3073 were charged off. On the other hand, 60-month loans had 10,421 observations with a lower success rate of fully paid loans. 8054 (77.3%) were fully paid off and 2367 were charged off. This shows us that the shorter loan terms are associated with a higher possibility of successful repayment.

	A	B	C	D
1	Term	Loan Status		
2			0	1 Grand Total
3	0	2367	8054	10421
4	1	3073	25214	28287
5	Grand Total	5440	33268	38708

The second pivot table I did was Interest Rate and Loan Status.

First the calculated average interest rate for the whole dataset was: 12.06%. Loans that are Charged Off (0) have a much larger average interest rate of 13.90%, while Fully Paid (1) loans have an average of 11.76%. This difference of over 2% suggests that higher interest rates are a primary indicator of risk in this dataset. This implies that higher borrowing costs make repayment more difficult, and riskier.

	A	B
1		Average of Interest_rate_cleaned
2	0	0.1390
3	1	0.1176
4	Grand Total	0.1206

Correlations:

	A	B	C	D
1		Loan_status_cleaned	Annual_income_cleaned	Loan_amount_cleaned
2	Loan_status_cleaned	1		
3	Annual_income_cleaned	0.06522608	1	
4	Loan_amount_cleaned	-0.039402519	0.426845872	1
5				

First correlation is Loan Status and Annual Income.

$r=0.06522$ so there is a positive correlation between annual income and loan status. This means as income rises the success rate of a loan being fully paid increases also. This suggests that people with a higher average income usually leads to a better loan outcome.

Second correlation is Loan Status and Loan Amount.

$r= -0.0394$ so there is a negative correlation between loan amount and loan status. This means as the amount borrowed increases the probability of a loan being fully paid decreases. This shows that the larger loans are usually riskier to pay off.

Regression:

For the final analysis I have conducted a regression analysis.

	A	B	C	D	E	F	G	H	I
1	SUMMARY OUTPUT								
2									
3	<i>Regression Statistics</i>								
4	Multiple R	0.2286							
5	R Square	0.0522							
6	Adjusted R Square	0.0521							
7	Standard Error	0.3384							
8	Observations	38708.0000							
9									
10	<i>ANOVA</i>								
11		<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>			
12	Regression	5	244.2929	48.8586	426.7324	0			
13	Residual	38702	4431.1726	0.1145					
14	Total	38707	4675.4655						
15									
16		<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>	<i>Upper 95%</i>	<i>Lower 95.0%</i>	<i>Upper 95.0%</i>
17	Intercept	0.4095	0.0369	11.1128	0.0000	0.3373	0.4818	0.3373	0.4818
18	Annual_income_cleaned	0.0579	0.0035	16.5147	0.0000	0.0510	0.0647	0.0510	0.0647
19	Loan_amount_cleaned	-0.0032	0.0028	-1.1606	0.2458	-0.0086	0.0022	-0.0086	0.0022
20	Interest_rate_cleaned	-1.5828	0.0524	-30.1857	0.0000	-1.6855	-1.4800	-1.6855	-1.4800
21	Emp_lenght_cleaned	-0.0021	0.0005	-4.1596	0.0000	-0.0030	-0.0011	-0.0030	-0.0011
22	Term_cleaned	0.0612	0.0045	13.5640	0.0000	0.0524	0.0701	0.0524	0.0701
23									

The regression resulted in an R-square of 0.0521 indicating that the selected independent variables explain around 5.21% of the loan repayment status. This might appear low but for a high scale financial dataset it should be significant.

Predictors ($P < 0.05$)

Annual Income: Coefficient = 0.0579 and P-Value= 0.000. As expected, higher income is a significantly positive predictor for successful repayment.

Loan Amount: Coefficient = -0.0032 and P-Value=0.2458. Since the p-value is higher than 0.05 this predictor is not statistically significant. This means the size of the loan does not matter as much as the other variables.

Interest Rate: Coefficient = -1.5828 and P-Value=0.000. This one is the most important predictor. As interest rates rise the probability of the loan being fully paid decreases.

Emp Length: Coefficient = -0.0021 and P-Value=0.000. This one is significant but shows as the years increase the probability of repaying decreases. This might be because people take larger loans if they feel more comfortable.

Term: Coefficient = 0.0612 and P-Value= 0.000. Shorter terms (36 months) have a higher success rate of repayment than longer (60 months) ones.

This model proves that interest rates and annual income are the most important variables out of these 5 for loan repayment. To minimize the risk of a borrower the model suggests prioritizing higher income individuals with short-term loans and competitive interest rates.

Section 5. Conclusions & Recommendations

In conclusion, the analysis of the loan dataset reveals insights into the drivers of repayment success, showing that loan performance depends on both the borrower's ability and loan conditions. Most significantly, the interest rate was a highly influential element affecting the outcome of the loan. The coefficient indicates that higher interest rates imply a significantly lower probability that a loan will be fully repaid, making it the most important predictor of risk in this model. This is supported by data showing that charged-off loans carried an average interest rate of 13.90%, while fully paid loans averaged only 11.76%. On the other hand, annual income became an equally important factor to support lending stability. The positive relationship between income and repayment means that higher-income borrowers have the financial ability to manage their loans. Besides, the term duration of loans matters too, as shown by the high success rates (89.1% in the case of 36 months and 77.3% when loans lasted 60 months). Surprisingly, what used to seem like a significant factor, such as loan amount, appears to be statistically insignificant.

To ensure better results in the future regarding loan approvals, it is advisable for the lender to concentrate more on selecting borrowers with higher incomes and making short-term loans at favorable interest rates to them. Taking into consideration the conclusions derived from the results, the most efficient strategy will be that which entails concentrating on offering 36-month loans and valuing annual income higher than job tenure and loan amount.